LAMP EUAs for SARS-CoV2

EUA Date	Test	Diagnostic	Target	IC	Detection	Extraction	Amplification	LOD spike	LOD GE/ 3ml swab
Nov-17	Lucira (At Home POC)	Lucira COVID-19 All-In-One Test Kit	N, N	External + IC	Colorimetric	~Direct	Lucira device	Inactivated virus	2,700
Oct-05	Seasun (2 duplex)	AQ-TOP COVID-19 Rapid Detection Kit PLUS	orf1ab, N	Human RNaseP	PNA Probe	Manual (Qiagen_60704 or Seasun_SS-1300) or Automated (Panagene_PNAK-1001 on PanaMax48)	BioRad_CFX96 or ABI7500	NCCP_43326 genomic RNA	3000
Sep-01	Detectachem	MobileDetect Bio BCC19 (MD-Bio BCC19) Test Kit	N, E	only external controls	Colorimetric	Direct (1µl of transport media into LAMP)	heat block or qPCR (MD-Bio heater or BioRad_T100 or AB_Veriti or ...)	Twist_MT007544.1 synthetic gRNA	225,000
Aug-31	Mammoth	SARS-CoV-2 DETECTR Reagent Kit	N	Human RNaseP	CRISPR Probe	Automated (Qiagen_955134 on Qiagen_EZ1AdvancedBenchtop)	ABI7500	SeraCare_AccuPlex_0505-0168 IVT encapsulated RNA	60,000
Aug-13	Pro-Lab	Pro-AmpRT SARS-CoV-2 Test	RdRP	only external controls	Probe	Direct (swab into 0.1ml) or kit (swab into 1ml then Pro-Lab_PLM-2000)	Optigene_GenieHT	BEI_NR-52287 inactivated virus	125*
Jul-09	UCSF/Mammoth	SARS-CoV-2 RNA DETECTR Assay	N	Human RNaseP	CRISPR Probe	automated (Qiagen_955134 on Qiagen_EZ1)	ABI7500	SeraCare_AccuPlex_0505-0126 (lot 10480311) IVT RNA	60,000
May-21	Seasun (1 duplex)	AQ-TOP COVID-19 Rapid Detection Kit	orf1ab	Human RNaseP	PNA Probe	manual (Qiagen_60704)	BioRad_CFX96 or ABI7500	NCCP_43326 gRNA	21,000
May-18/ Aug31	Color	Color Genomics SARS-CoV-2 RT-LAMP Diagnostic Assay	N, E, nsp3	Human RNaseP	Colorimetric	automated (PerkinElmer_CMG-1033 on PerkinElmer_Chemagic360)	plate reader (Biotek_NEO2), Hamilton_Star	ATCC_VR-1986D (gRNA)	2,250
May-06	Sherlock BioSci	Sherlock CRISPR SARS-CoV-2 Kit	orf1ab, N	Human RNaseP	CRISPR Probe	Manual (ThermoFisher_12280050)	heat block or qPCR, Plate reader (BioTek_NEO2)	gRNA	20,250
Apr-10	Atila BioSystems	iAMP COVID-19 Detection Kit	orf1ab, N	human GAPDH	Probe	Direct (swab into 350µl, 15min RT then 3µl to LAMP)	BioRad_CFX96 or ABI7500 or Roche_LightCycler480II or Atila_PG9600	SeraCare_AccuPlex_0505-0129 pseudovirus (recombinant alphavirus)	3,500*

Just in 2020!
from Matt McFarlane
(twitter @mattmcfar)

contact@floodlamp.bio

Resources

[ASU Testing Commons]

FIND

OTC LAMP EUA – Performance Summary

Company	EUA LOD (cp/swab)	EUA Clin PPA (sensitivity)	EUA Clin NPA (specificity)	Intended Use
Lucira	2,700	92%	98%	OTC, asymptomatic
Detect	313	91%	98%	OTC, asymp serial
Uh-Oh Labs	400	88%	100%	POC, persons suspected of COVID
Aptitude - Swab	2,000	96%	98%	OTC, asymptomatic
Aptitude - Saliva	20 cp/uL	90%	99%	OTC, asymptomatic
Cue (? isothermal)	20			OTC, asymptomatic

OTC LAMP EUA - Platform Comparison

Platform	Instrument cost	Per-test cost	LOD (Viral Genomic RNA)	Accuracy to PCR	Total test time	Incubation time
Uh-Oh Labs	$200	$25	400 copies/swab	96%	17 - 45 mins	12 - 40 mins
Cue	$159	$52	20 copies/swab 1.3 copies/uL	98%	22 mins	20 mins
Detect	$39	$49	313 copies/swab <1 copy/uL	97%	70 mins	55 mins + 10 mins read time
Lucira	NA	$75	2700 copies/swab	98%	13 - 32 mins	11 - 30 mins

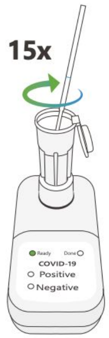
Lucira

Lucira Test - At-Home/OTC

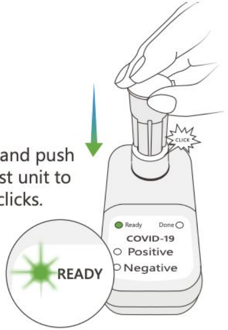
Intended Use

The Lucira CHECK✓IT COVID-19 Test Kit is a single-use test kit intended for the qualitative detection of nucleic acid from the novel coronavirus SARS-CoV-2 that causes COVID-19. This test is for nonprescription home use with self-collected anterior nasal (nasal) swab specimens in individuals aged 14 years and older (self-collected) or individuals ≥2 years (collected by an adult) with or without symptoms or other epidemiological reasons to suspect COVID-19.

3 Stir Swab and Run Test



- Insert swab into the sample vial until it **touches the bottom.**
- Mix sample by **stirring around the sample vial 15 times.**
- Discard swab.



- Snap cap closed and push vial down into test unit to start test until it clicks.
- Ready light will start **blinking** when test is running.

If Ready light is not blinking within 5 seconds, use palm of your hand to push down more firmly to start test.

- Do not move test unit once the test has started running.

Wait 30 minutes.

Lucira Test - Technology

Lucira LOD: 2,700 cp/swab

Lucira EUA IFU

1) Limit of Detection (LoD) - Analytical Sensitivity

Quantified heat-inactivated SARS-CoV-2 virus was serially diluted in Natural Nasal Swab Matrix (NNSM), 35 µL was pipetted onto a fresh, unused nasal swab, and run on two device lots. The LoD for the device was determined by testing three (3) target concentrations on each lot of devices. For each lot, each concentration was tested in replicates of seven (7) devices by three (3) unique operators, for a total of 21 replicates per concentration. The LoD for each lot was separately determined as the lowest concentration of genome copy equivalents per swab that yielded greater than 95% positive results. The preliminary LoD for the device was defined as the highest LoD of the two lots.

Table 1. LOD Determination Results

Genome equivalent / Swab (per reaction)	Positive/Total Valid			Percent Positive		
	Preliminary Lot 1	Preliminary Lot 2	Confirmatory	Preliminary Lot 1	Preliminary Lot 2	Confirmatory
2700*	20/21	20/21	20/20	95.2%	95.2%	100%
1350	18/21	20/21	--	85.7%	95.2%	--
675	16/21	15/21	--	76.2%	71.4%	--

*2700 cp/swab is determined as LOD and is equivalent to 900 cp/mL of VTM assuming 100% elution of the swab in 3 mL of VTM.

The LoD for Lot 1 was determined to be 2700 copies per swab, while Lot 2 was 1350 genome equivalents per swab. As such, 2700 copies per swab was reported as the preliminary LoD.

The LoD was confirmed by testing 20 replicates at the preliminary LoD concentration on a single lot. All 20 of 20 devices at this concentration were positive.

Lucira Clin Eval: 92% PPA, 98% NPA

Lucira EUA IFU

Table 12. Community Testing Clinical Study Summary

Subject Type	Lucira Community Testing Study Summary Clinical Performance Vs High Sensitivity FDA EUA SARS-CoV-2 PCR Test	
	PPA	NPA
Symptomatic	94% (48/51)	98% (49/50)
Asymptomatic	90% (73/81)	98% (218/222)
Total	92% (121/132)	98% (267/272)

*Seven (7) of the 11 discrepant samples had high Ct values (>37.5) when tested by the comparator assay.

Table 13. Community Testing Study Summary Results

Lucira Community Testing Studies	A High Sensitivity FDA Authorized SARS-CoV-2 PCR Test		
	Positive	Negative	Total
Positive	121	5	126
Negative	11	267	278
Total	132	272	404

Positive Percent Agreement (PPA) 91.7% (95%CI: 85.6%-95.8%)

Negative Percent Agreement (NPA) 98.2% (95%CI: 95.8%-99.4%)

7 invalid result (1.73% invalid rate); 2 retests (0.49% retest rate)

*Seven (7) of the 11 discrepant samples had high Ct values (>37.5) when tested by the comparator assay.

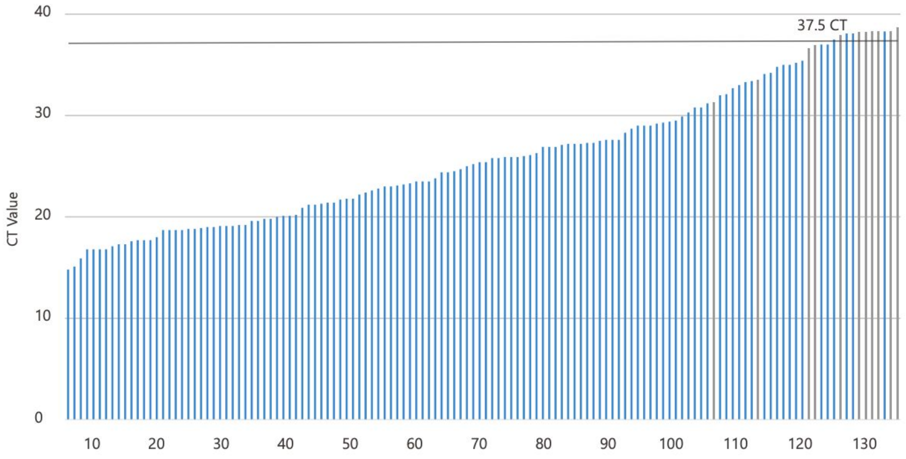
Detect

Lucira Test – Community Study Plot

Lucira EUA IFU - Home Insert

Table 13. Lucira vs. a High Sensitivity Molecular FDA Authorized SARS-CoV-2 PCR Assay Positive Percent Agreement Summary

92% PPA across all samples (121/132*)



Samples shown by decreasing amount of virus

The # of cycles (CTs) required to detect virus increases when the amount of virus in the sample is low

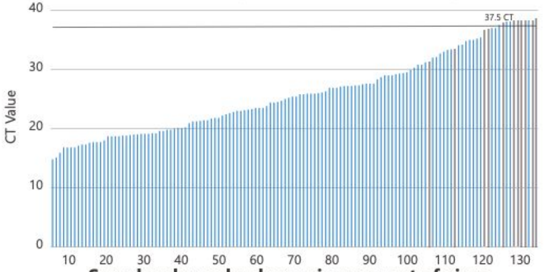
*Seven (7) of the 11 discrepant samples had high Ct values (>37.5) when tested by the comparator assay.

How accurate is this test?

Lucira's CHECK✓IT COVID-19 Test Kit is a molecular in vitro diagnostic test that has an analytical sensitivity, or ability to detect the SARS-CoV-2 virus, that is comparable to some of the best molecular PCR tests performed in clinical settings and high complexity labs.

In two Community Testing Studies which included 404 individuals with and without COVID-19 symptoms, the Lucira test achieved 98% (267/272) negative percent agreement (NPA) when compared to a FDA authorized known high sensitivity SARS-CoV-2 PCR test. Positive percent agreement (PPA) among symptomatic individuals was 94% and 90% in asymptomatic. Total PPA was 92% (121/132) across all samples and included 10 samples with very low levels of virus >37.5 Ct as shown below.

Lucira Community Testing Study
Positive Percent Agreement (PPA) Summary



Samples shown by decreasing amount of virus

The # of cycles (CTs) required to detect virus increases when the amount of virus in the sample is low

Detect Test

Detect

Detect Covid-19 Test™

Covid-19 Molecular Home Test

Self-collect your sample

Prepare your sample

Mix the sample with our proprietary reagent bead

Use the reusable Detect Hub™ to process sample in 55 min

Read results in 10 min

Detect Test - At-Home/OTC, Asymptomatic Serial

1. Intended Use

The Detect Covid-19 Test™ (the Detect test) is a molecular *in vitro* diagnostic test for the qualitative detection of nucleic acid from the novel coronavirus SARS-CoV-2 that causes Covid-19.

This test is authorized for non-prescription home use with self-collected anterior nasal (nasal) swab samples from individuals aged 14 years or older suspected of Covid-19. This test is also authorized for non-prescription home use with adult-collected anterior nasal swab samples from individuals aged 2 years or older suspected of Covid-19.

This test is also authorized for non-prescription home use with self-collected anterior nasal (nasal) swab samples from individuals aged 14 years or older, or adult collected anterior nasal swab samples from individuals aged 2 years or older, without symptoms or other epidemiological reasons to suspect Covid-19 when tested twice over three days with at least 24 hours (and no more than 48 hours) between tests.

In asymptomatic individuals (those without Covid-19 symptoms), the Detect Covid-19 Test should be used as a serial test.

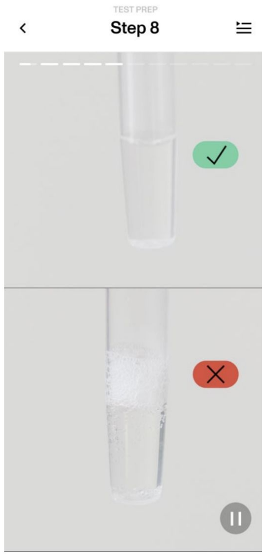
What is serial testing?

Serial testing involves testing the same person multiple times within a few days. Such testing for Covid-19 increases the chances of identifying infections earlier and should be used for people who are not exhibiting any symptoms.

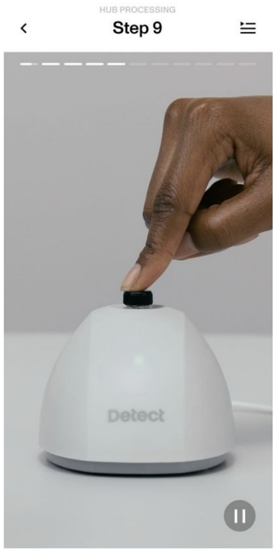
How do I use the Detect Covid-19 Test for serial testing?

If you're asymptomatic and not exposed to Covid-19 and your first test is negative, you take a second test after at least 24 hours but within 48 hours. If your first test is positive, then you are likely to have Covid-19 currently and should consult with a healthcare provider without waiting to use the second test. If only your second test is positive, then you are also likely to have Covid-19 and should consult with a healthcare provider.

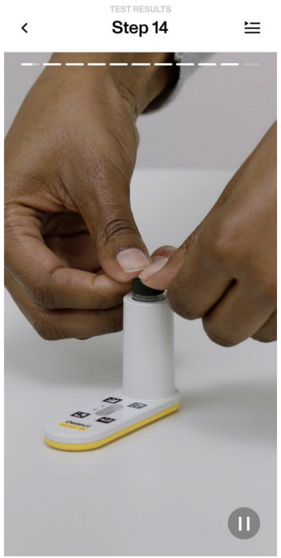
Detect Workflow



REMINDER: Liquid clinging to tube walls or cap may result in an invalid test.

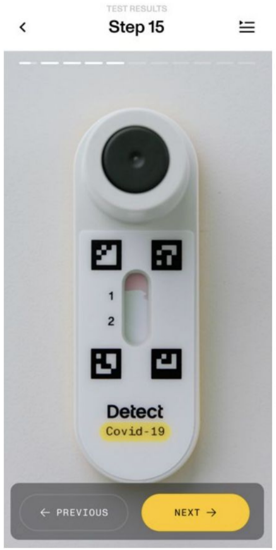


REMINDER: Do not remove the Test Tube from the Hub until it is time to use it.



REMINDER: Keep pressing until the tube is completely flush with the lip of the Reader chimney. Liquid should begin to flow through the Reader and should be visible within the Reader's window in about 10 seconds. If there's no flow, firmly tap the Reader against a hard surface 3 times to help trigger flow.

Detect Readout



- The red lines will take approximately 10 minutes to fully develop. Line darkness and fullness may vary.

- As shown below on the right, lines can be faint or incomplete.

Example result shown is positive.

Detect LOD: 313 cp/swab

Detect EUA IFU

11.1 Analytical Sensitivity (Limit of Detection)

The Limit of Detection (LoD) is the lowest amount of virus that can be determined in 95% of samples and was determined by testing the Detect test's analytical sensitivity with heat-inactivated SARS-CoV-2 BEI, USA-WA1/2020) in pooled nasal matrix. Over two lots of tests, the LoD of the Detect Covid-19 Test was determined to be 313 copies per swab, which is equivalent to 800 copies per mL if all virus is transferred from the swab to the buffer.

Viral Load (genomic copies/swab)	Lot 1 SARS-CoV-2 Detection Rate	% Detected	Lot 2 SARS-CoV-2 Detection Rate	% Detected
313	20/20	100%	20/20	100%
156	20/20	100%	18/20	90%
78	15/20	75%	14/20	70%

Detect Clin Eval: 91% PPA, 98% NPA

Detect EUA IFU

11.5 Clinical Evaluation

A prospective, multi-center clinical study was conducted in the United States in subject's homes or a simulated home environment. Testing was performed by the untrained subject or the subject's parent/guardian for children under 14. The study enrolled symptomatic subjects and asymptomatic subjects with a recent exposure, each of whom self-collected two nasal swab samples. For each subject, one swab was collected and sent to a reference laboratory and tested using a high sensitivity FDA-authorized SARS-CoV-2 RT-PCR test by trained laboratory personnel as a comparator, while the other swab was run through the Detect test by the untrained subject or parent/guardian. Comparing the Detect test's results to those produced by the high sensitivity FDA-authorized SARS-CoV-2 RT-PCR test, the Positive Percent Agreement (PPA) was 90.9.% (30/33) and the Negative Percent Agreement (NPA) was 97.5% (77/79). Both apparent false positives were due to subject's misinterpreting the test result; the Detect App has since been modified to significantly reduce the frequency of this user error.

Clinical Study Results Summary

	FDA-authorized SARS-CoV-2 RT-PCR Assay	
Detect	Positive	Negative
Positive	30	2*
Negative	3**	77

*Both of the apparent false positive results were due to subject misinterpretation using an older version of the Detect App. A modified version of the Detect App has since been developed and shown to be effective in reducing the frequency of subjects misinterpreting the test result.
**One of the three apparent false negative samples gave a negative result when tested with a second highly sensitive EUA SARS-CoV-2 RT-PCR assay.

Positive Percent Agreement (PPA): 90.9% (30/33), (95% CI: 76.4%-96.9%)
Negative Percent Agreement (NPA): 97.5% (77/79), (95% CI: 91.2%-99.3%)

A second evaluation was conducted testing the ability of untrained lay users to correctly identify and interpret near-cutoff positive samples with the Detect test. Twelve subjects were each given three blinded contrived swab samples, a mix of negative and low-positive (1.9x LoD) to test. Comparing their results to those expected, the PPA was 94.4% (17/18) and the NPA was 100.0% (18/18), demonstrating that in the hands of lay users the Detect Covid-19 Test can reliably identify low-positive samples.

Near-Cutoff Results Summary

	Contrived Sample Expected Result	
Detect	Positive	Negative
Positive	17	0
Negative	1	18

Positive Percent Agreement (PPA): 94.4% (17/18), (95% CI: 74.2%-99.0%)
Negative Percent Agreement (NPA): 100.0% (18/18), (95% CI: 82.4%-100.0%)

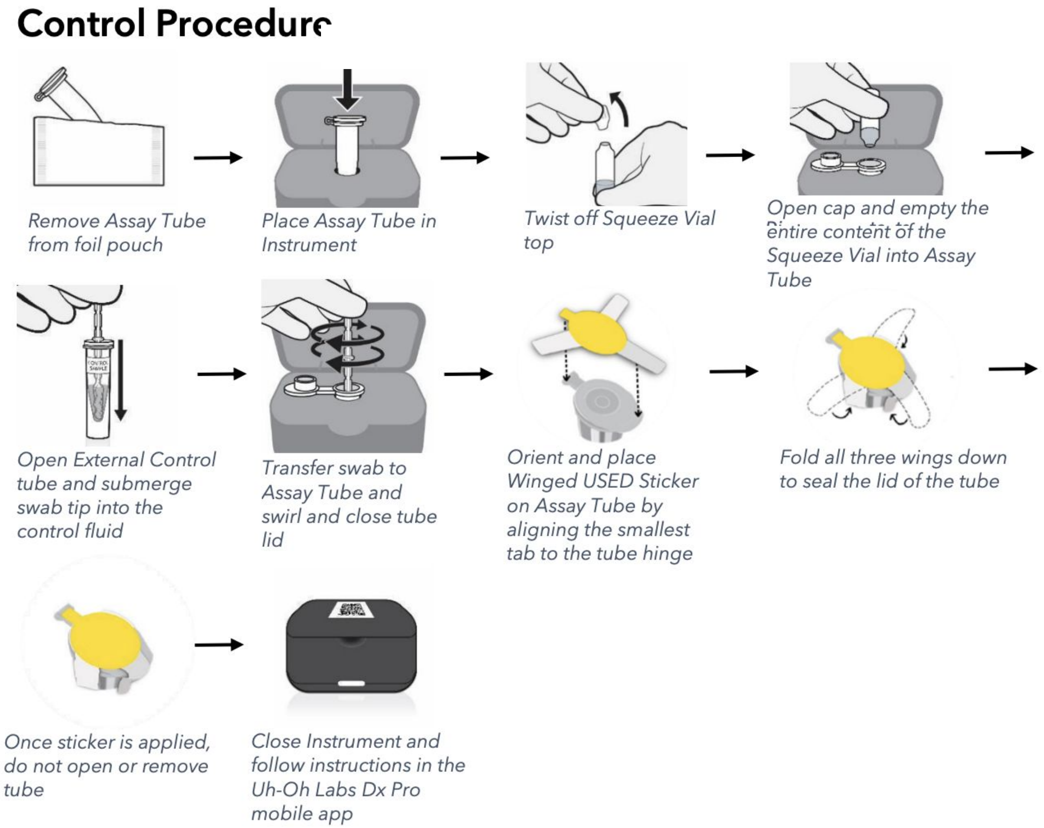
Uh-Oh Labs

Uh-Oh Labs COVID Test - POC/CLIA Waived

INTENDED USE

The UOL COVID-19 Test is a real-time RT-LAMP test intended for the qualitative detection of nucleic acid from SARS-CoV-2 in anterior nasal swab specimens from individuals suspected of COVID-19 by a healthcare provider. Testing is limited to laboratories certified under the Clinical Laboratory Improvement Amendments of 1988 (CLIA), 42 U.S.C. §263a, that meet requirements to perform high, moderate, or waived complexity tests. The UOL COVID-19 Test is authorized for use at the Point of Care (PoC), i.e., in patient care settings operating under a CLIA Certificate of Waiver, Certificate of Compliance, or Certificate of Accreditation.

Uh-Oh Labs COVID Test



Uh-Oh Labs LOD: 400 cp/swab

Uh-Oh Labs EUA IFU

Analytical Sensitivity (Limit of Detection, LoD) - Heat Inactivated SARS-CoV-2 Virus

The LoD studies establish the lowest SARS-CoV-2 viral concentration (copies per swab) in a specimen that can be detected by the UOL COVID-19 Test at least 95% of the time. The initial LoD was determined by testing six input concentrations (1600, 800, 400, 200, 100 and 50 copies per swab) of heat-inactivated SARS-CoV-2 virus spiked into negative anterior nares swab samples in rehydration buffer. The preliminary LoD estimate of 400 copies per swab was confirmed by running 20 additional copies (Table 2).

Table 2. LoD Study results.

Genome copies/swab	SARS-CoV-2 detected/tested	% Detected
1600	3/3	100%
800	3/3	100%
400 (initial)	3/3	100%
400 (confirmation)	**20/20**	**100%**
200	5/6	83%
100	5/6	83%
50	1/3	33%

Uh-Oh Labs Clin Eval: 88% PPA 100% NPA

Uh-Oh Labs EUA IFU

Clinical Evaluation

The ability of untrained operators to successfully run the UOL COVID-19 Test was evaluated using prospectively collected samples from three separate and geographically diverse point of care sites in the U.S. that reflected a spectrum of point of care uses. The study was IRB approved and patients were consented according to the protocol. Patients presenting to the 3 point of care settings with signs and/or symptoms of COVID-19 or possible exposure to COVID-19 individuals were tested using the UOL COVID-19 Test. A total of 207 individuals were tested with the UOL COVID-19 Test and the results compared to those generated on a paired NP swab tested with an EUA authorized high-sensitivity RT-PCR assay as the comparator assay. The positive percent agreement (PPA) was 87.7% and the negative percent agreement (NPA) was 100% (Table 1). The significant majority (7/8) of the false negative results were from samples with high Ct values of the comparator (Ct ≥33.4 or not detected) when tested by the comparator assay.

Table 1. UOL COVID-19 Test Clinical Performance

UOL COVID-19 Test	High-sensitivity FDA authorized RT-PCR assay		
	Positive	**Negative**	**Total**
Uh-Oh Labs Positive	57	0	57
Uh-Oh Labs Negative	8*	142	150
Total	65	142	207
Positive Percent Agreement (PPA)	87.7% (95% CI = 77.2-94.5%)		
Negative Percent Agreement (NPA)	100% (95% CI = 97.4 to 100%)		
Overall Agreement (OA)	96.1% (95% CI = 92.5 to 98.3%)		

** Six (6) of the 8 discordant results had Cts above the reported mean Ct at the LoD of the comparator test. Three (3) of the 8 discordant results were detected for a single target only on the comparator. Of those, two samples were 'presumptive positive' because the comparator only detected target 2 with Cts of 39.4 and 35.6 (both below the mean Ct at the comparator LoD) respectively.*

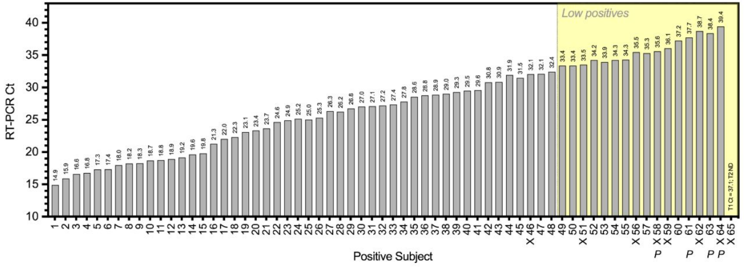
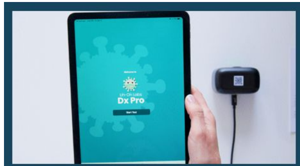


Figure 1. The UOL COVID-19 Test detected 97.9% of SARS-CoV-2 positive samples, excluding those with very low levels of virus. This graph shows the measured Ct values for Target 2 by the EUA authorized high-sensitivity RT-PCR comparator for each evaluable subject in the clinical study. The Ct values are written above each bar. UOL COVID-19 Test false negatives are marked with an X below the subject number, and comparator presumptive positives are marked with a P. The yellow shaded area denotes the range at which samples are considered low positives (Ct ≥ 33.4).

Uh-Oh Labs – Test

Copy from website: “The UOL COVID-19 Test is an FDA-authorized, PCR-quality COVID test system that returns results in as fast as 12 minutes with a 96.1% Accuracy compared to high-sensitivity PCR - all for a lower cost than other CLIA-waived COVID NAATs.”



Start

Open the DxPro app, scan your Instrument, scan your Kit, and register the patient

Squeeze

Open the Test Kit, insert Assay Tube, and squeeze all of the Vial solution into the Assay tube

Swab

Use the sterile nasal swab and collect a sample from both nostrils by making 5 big circles inside the nose

Stir

Use the sterile nasal swab and collect a sample from both nostrils by making 5 big circles inside the nose

Submit

Close the Assay Tube, seal it with the included sticker, and submit the test - that's it!

Uh-Oh Labs - Tech Specs

- 2 Color Fluorescence
- Individual Reader

TECHNICAL SPECIFICATIONS

UOL COVID-19 Instrument

Part number	UOL002
Product description	UOL COVID-19 Instrument
Sample capacity (wells)	1 well per instrument
Reaction volume	150 microliters
Excitation source	Blue and green LEDs
Detection channels	FAM, HEX
Multiplexing	2 channels
Thermal element	Resistive cartridge heater
Temperature accuracy	+/- 1°C, maximum +/- 2.5°C
Custom dye/chemistry	Loop-de-Loop™ RT-LAMP
Connectivity	Bluetooth
Power Supply Input	100-240V AC 50/60Hz
Power Supply Output	5VDC, 2A, USB-C
Operating and storage conditions	Temperature: +2°C to +30°C Relative humidity: < 95% RH non-condensing Atmospheric pressure: 70 kPa to 106 kPa
Other environmental requirements	Keep out of direct sunlight, keep dry (regular cleaning per IFU is OK), use upright
Transit conditions	Temperature: -18°C to +60°C Relative humidity: < 90% RH Atmospheric pressure: 59 kPa to 106 kPa
Protection against electric shock	Class II
Dimensions	60 mm x 100 mm x 65 mm
Weight	106 grams

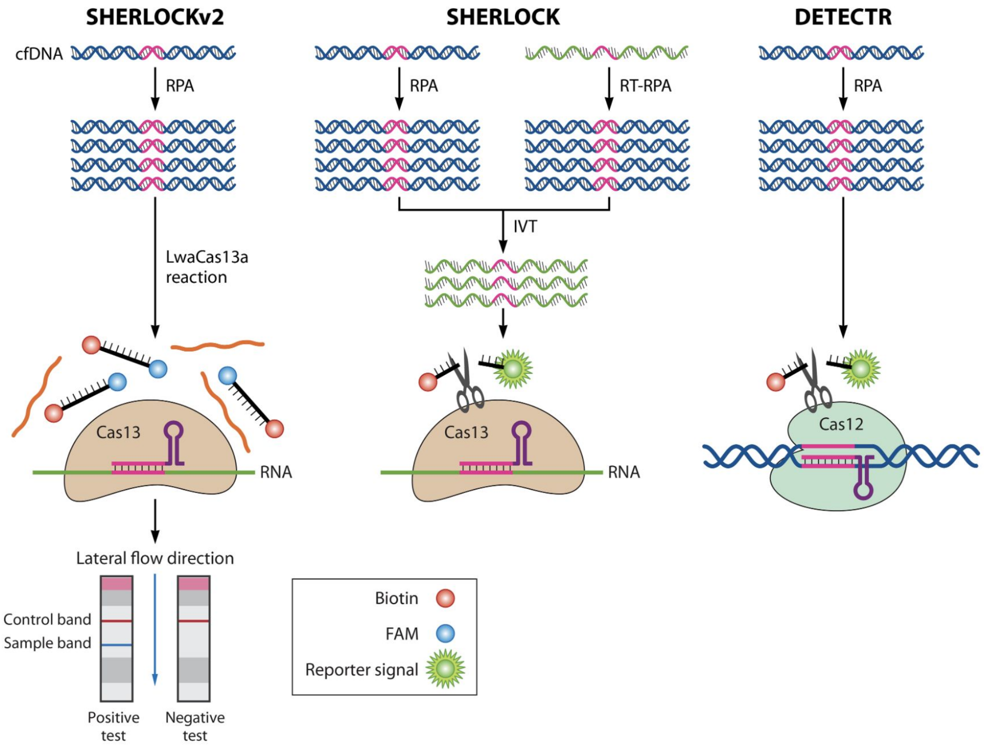
UOL COVID-19 Test Kit

Operating and storage conditions	Temperature: +2°C to +30°C Relative humidity: < 95% RH non-condensing Atmospheric pressure: 70 kPa to 106 kPa
Transit conditions	Temperature: -18°C to +60°C Relative humidity: < 90% RH Atmospheric pressure: 59 kPa to 106 kPa

Sherlock

https://www.360dx.com/molecular-diagnostics/sherlock-biosciences-expects-sense-biodetection-buy-accelerate#.Y-1P-OzMJqs - GDoc version

How Sherlock works video clip

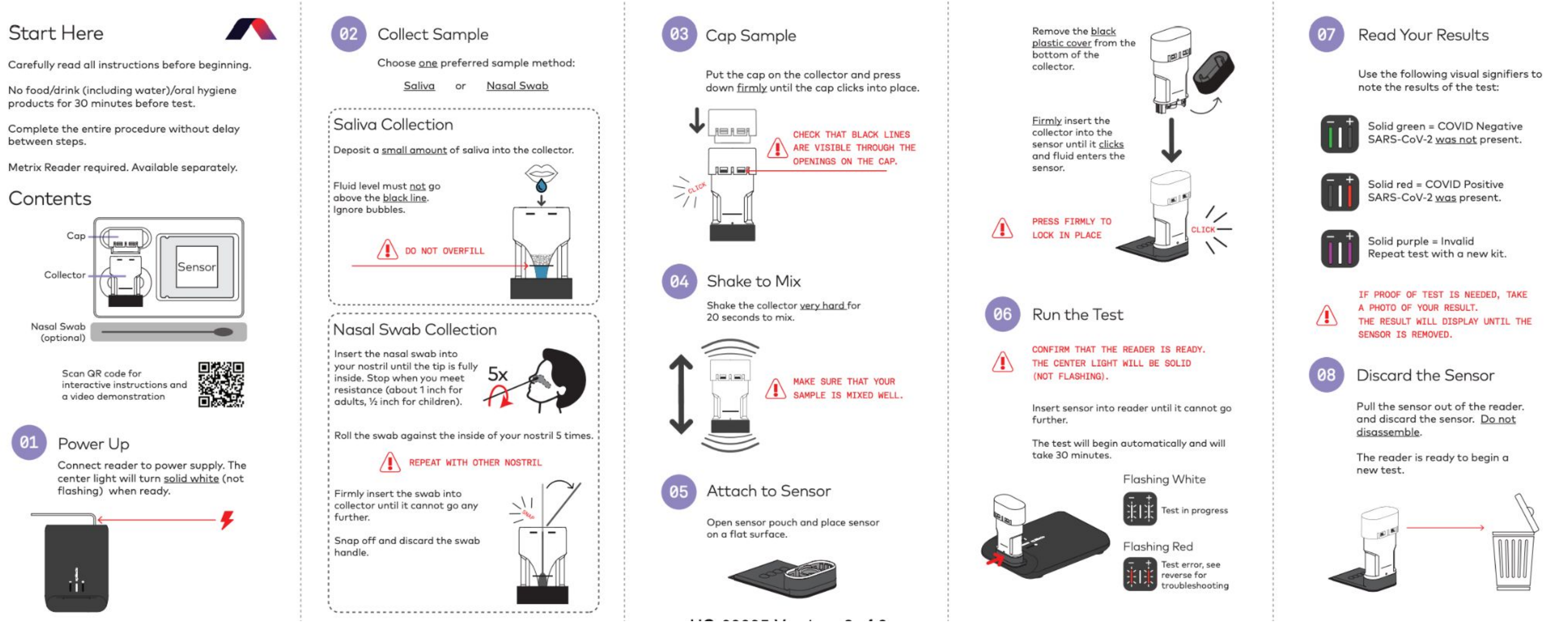


Aptitude Medical

Aptitude Medical - Metix COVID Test

- OTC at-home use
- Swab and Saliva

self collected from any individual aged 14 years or older, or adult-collected from any individual aged 2 years or older, including individuals without symptoms or other epidemiological reasons to suspect COVID-19.



Aptitude Medical – IFU Doesn't Disclose LAMP

This test utilizes nucleic acid amplification technology, similar to PCR, for the detection of SARS-CoV-2. SARS-CoV-2 viral RNA is generally detectable in anterior nasal (nares) swab and saliva samples during the acute phase of infection.

3. Principles of the Procedure

The Metrix COVID-19 Test is a compact nucleic acid amplification test (NAAT) that detects the genetic material of SARS-CoV-2 in unprocessed saliva or nasal swab specimens using an isothermal molecular amplification reaction that is an equivalent alternative to polymerase chain reaction (PCR). The Metrix COVID-19 test utilizes a multi-gene amplification method utilizing primers that target both the nucleocapsid (N) and open reading frame (ORF-1) genes of SARS-CoV-2. The Metrix COVID-19 Test also identifies nucleic acids from a human gene that serves as a control for proper assay execution, sample inhibition, amplification, and assay reagent function.

Aptitude LOD: 2,000 cp/swab 20 cp/uL saliva

Aptitude EUA IFU

10.1 Analytical Sensitivity (Limit of Detection)

The LoD studies for saliva and swab samples were conducted to determine the lowest detectable concentration of the Metrix COVID-19 Test at which 95% of all true positive replicates yielded a positive result in the test. LoDs were determined via a range-finding study followed by a limiting dilution study. Specific viral loads were set (i.e., genome equivalents, GE) using contrived positive samples created by serially diluting γ-irradiation-inactivated SARS-CoV-2 virus into pooled saliva or nasal matrix that was verified to be SARS-CoV-2-free with RT-PCR. For each sample type, the range-finding study was conducted to explore a wide range of viral concentrations to find a preliminary LoD at which 3/3 samples showed positive results in the Metrix COVID-19 Test. The preliminary LoD for each sample type was then used in the limiting dilution study to test positive samples at a narrower concentration range at 20 replicates per sample. The LoD was then confirmed across multiple lots of devices by repeating another 20 replicates at the LoD concentration in a second lot of devices (see tables below for each sample type). Replicate negative samples were also run and all showed negative results in the tests as expected.

The LoD of nasal swab samples was determined to be 2000 GE/swab, which is equivalent to 667 copies per mL of VTM assuming 100% elution of the swab in 3 mL of viral transport media (VTM). The LoD of saliva samples was determined to be 20 GE/μL of saliva.

Limit of Detection: Anterior Nasal (Nares) Swab Samples

Lot	Swab Viral Load (GE/swab)	#Positive/#Tested	% Positive
1	4000	20/20	100
1	2000	20/20	100
1	1000	14/20	70
2	2000	20/20	100

Limit of Detection: Saliva Samples

Lot	Saliva Viral Load (GE/μL)	#Positive/#Tested	% Positive
1	40	20/20	100
1	20	19/20	95
1	10	15/20	75
2	20	20/20	100

Aptitude Clin Eval: Swab 96% PPA, 98% NPA
Saliva 90% PPA, 99% NPA

Aptitude EUA IFU

Nasal Swab

Anterior Nasal (Nares) Swab Samples:
For anterior nasal (nares) swab samples, the Positive Percent Agreement (PPA) was 95.0% (38/40) and the Negative Percent Agreement (NPA) was 97.1% (34/35) in the symptomatic group. In the asymptomatic group,

the PPA was 100.0% (11/11) and the NPA was 98.6% (143/145). The PPA was 96.1% (49/51) and the NPA was 98.3% (177/180) in the combined total study population. There were 3 invalid tests (1.3%) and no canceled tests. The initially candidate invalid results were retested with concordant results. When excluding the single sample (Days Post Symptom Onset (DPSO) = 4) with a viral load below the comparator LoD, the Metrix COVID-19 Test correctly identified 49/50 positive nasal swab samples.

Subject Type	Metrix COVID-19 Test – Swab Assay vs. Comparator EUA SARS-CoV-2 RT-PCR Assay	
	PPA	NPA
Symptomatic	95.0% (38/40)	97.1% (34/35)
Asymptomatic	100.0% (11/11)	98.6% (143/145)
Total	**96.1% (49/51)**	**98.3% (177/180)**

Saliva

Saliva Samples:
Comparing the Metrix COVID-19 Test results, when using saliva samples, the symptomatic group PPA was 90.2% (37/41) and the NPA was 100.0% (35/35). The asymptomatic group PPA was 90.9% (10/11) and the NPA was 98.6% (143/145). For the combined study population, the PPA was 90.4% (47/52) and the NPA was 98.9% (178/180). There were 6 invalid tests (2.6%) and 2 canceled tests (0.9%) due to device errors requiring re-tests. Samples were retested per the IFU with a valid result.

Subject Type	Metrix COVID-19 Test – Saliva Assay vs. Comparator EUA SARS-CoV-2 RT-PCR Assay	
	PPA	NPA
Symptomatic	90.2% (37/41)	100.0% (35/35)
Asymptomatic	90.9% (10/11)	98.6% (143/145)
Total*	**90.4% (47/52)**	**98.9% (178/180)**

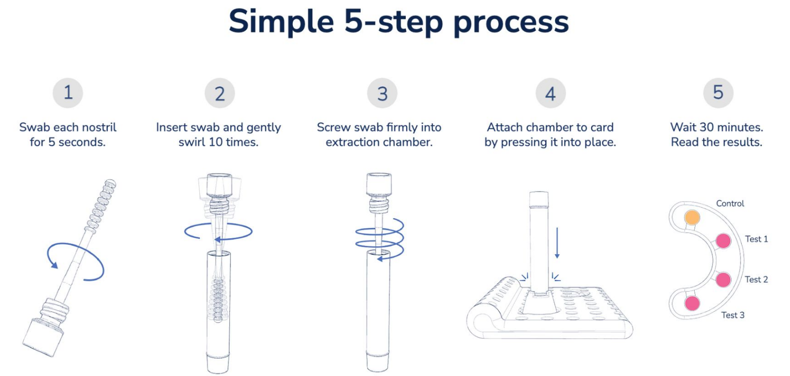
* The Metrix COVID-19 Test with saliva specimens was also evaluated against a saliva-based EUA RT-PCR test (SalivaDirect). 228/232 results were in agreement with the saliva-based EUA test, which is 3 fewer discordant results relative to the comparator that used NP samples (7 discordant results). The saliva-based EUA test characterized these 3 samples as negative while the NP comparator characterized them as positive. One of these 3 subjects was asymptomatic while the DPSO of the other 2 subjects were 4 and 7 days.

Domus

https://www.nature.com/articles/s41598-022-11144-5.pdf

former CEO of Quanterix

see gdoc



Midge Medical

[Genome Web article 9-13-22](#)

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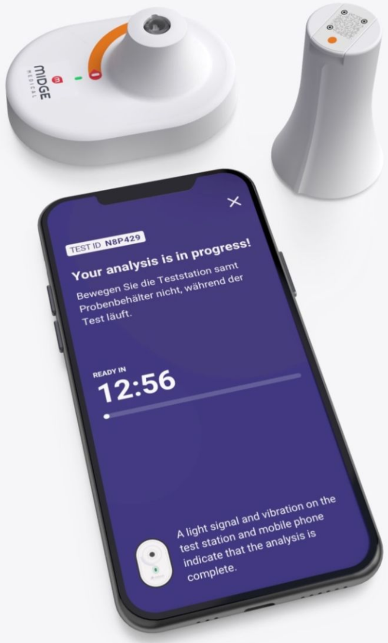
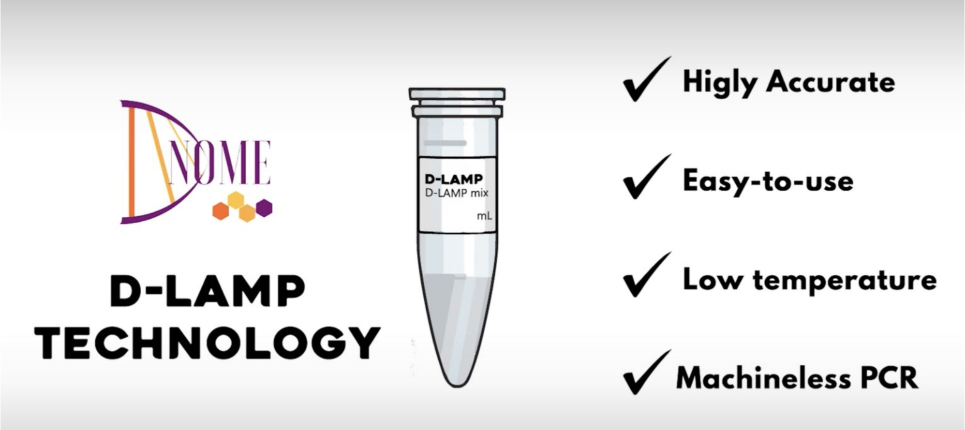
Anonymised data

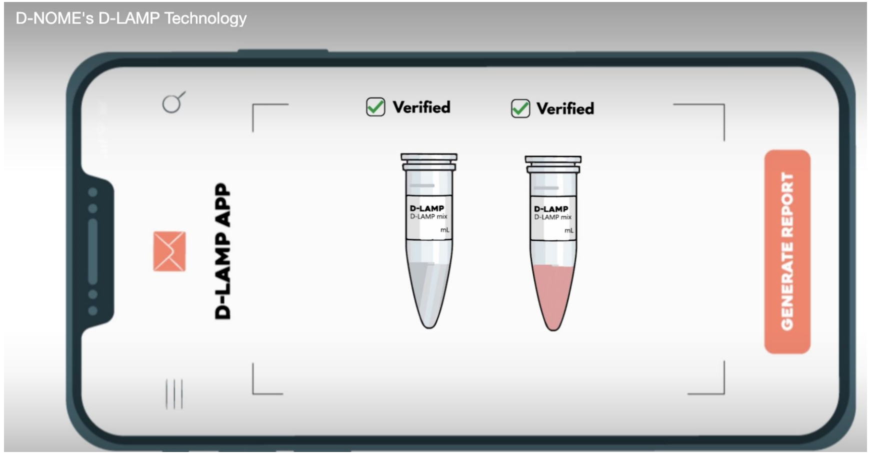
Through our specially developed IT platform, we can provide access to anonymized data for clinics and universities, as well as health authorities and scientists.

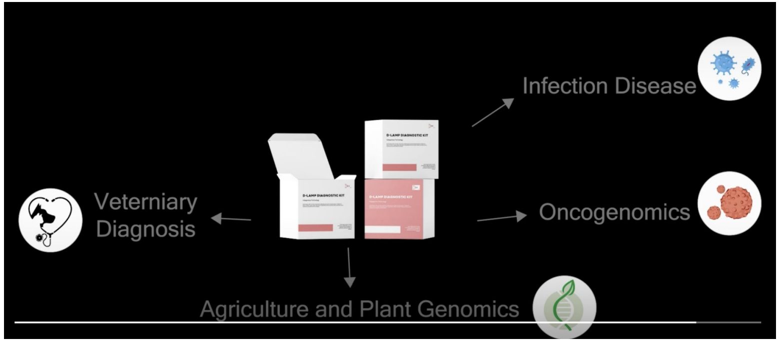


D-Nome - Balvi funded LAMP test in India

https://www.dnome.in/products

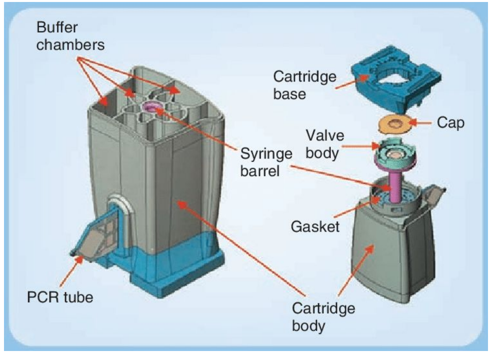


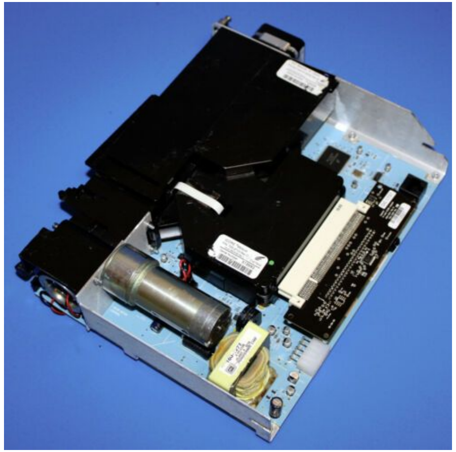




Cepheid – Legacy POC

acq Danaher
2016



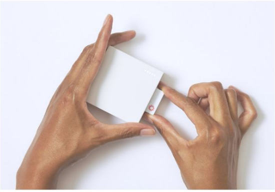


Cepheid - Omni

see gdoc 360Dx article

Time for $5 Coalition quickly applied public pressure — in the form of an open letter — to try to persuade Cepheid to reinstate the Omni project.

Cue



01

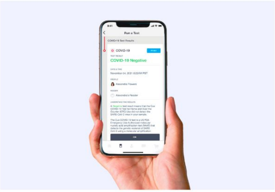
Insert your Cartridge into the Reader.

02

Use the Wand to collect a sample from the lower part of each nostril.

03

Insert the Wand into the Cartridge.



04

Get results from the Cue Health App on your mobile device in 20 minutes.

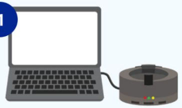
12-15 Molecular Diagnostics

presenting at Saliva Direct Conference
https://12-15mds.com/solutions/

How is this different from other COVID tests?

	VERALIZE	PCR-Based	LAMP-Based	Antigen
Accuracy	★★★★	★★★★★	★★★	★
Turn Around Time	7 minutes	Days	20 minutes	15 minutes
Sample Type	Saliva	NP/OP	Nasal	Nasal
Complexity	CLIA Waived	High	Moderate	CLIA Waived
Variant detection	YES	YES	YES	X
Multiplex capability	YES	X	X	X
Potential beyond COVID	YES	YES	YES	X
Suitable for home use	YES	X	YES	YES
Low cost to user	$18	X (>$100)	X ($90)	YES ($25)
Pre-symptomatic detection	YES	YES	YES	X
Platform technology	YES	X	X	X

How it Works

1


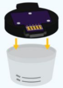
Plug the **Veralize Test Unit** to the wall outlet and connect it to the **laptop**. Start the **Veralize App**. Input the subject particulars.

2


Drool saliva into the **disposable cup**. The saliva level should be between the two marks.

3


Add the entire bottle of **nuclease-free** water to the disposable cup.

4


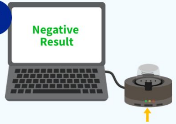
Close the **sensor lid** on top of the disposable cup.

5


Swirl the disposable cup for **5 seconds** to mix the contents.

6


Invert the disposable cup into the Test Unit until it locks.

7


Wait approximately 7 minutes. The result is displayed by the **3 LED indicators** as well as on the laptop.

8


Discard the disposable cup and sensor lid.

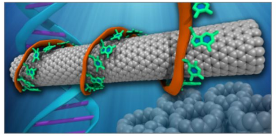
Veralize's Technology is Elegantly Simple

Veralize's patented technology uses state-of-the-art Graphene Carbon-Nanotubes that are mixed with two preserved regions of synthetic (single-stranded-DNA) COVID-19 genome.

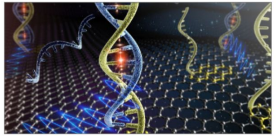
The nanotubes are an excellent support matrix to present the single-stranded-DNA (ss-DNA) probe fragments for hybridization with the test sample's ss-RNA.

Test sample's virus shell (if present) from fresh saliva samples will be broken down by our patent-pending process, thus releasing the ss-RNA within the shell.

Positive test occurs when the test sample's ss-RNA hybridizes with the probe's ss-DNA strands, producing an increased electric charge.



GCN support matrix

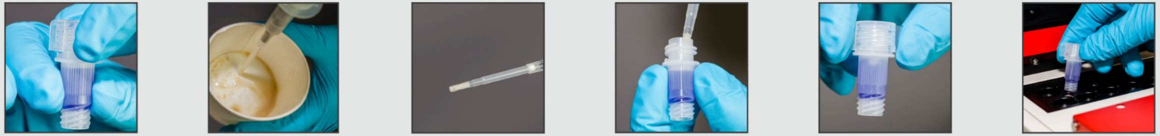


ss-RNA/ss-DNA hybridizing

Eiken – TB LAMP

TB-LAMP Workflow

1. Sample transfer and lysis

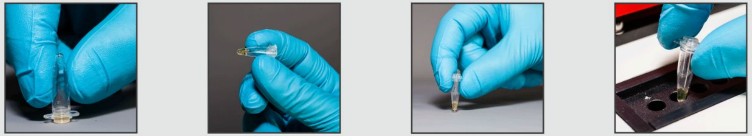


- Remove the cap to open the heating tube of the Loopamp™ PURE DNA Extraction Kit.
- Use the Pipette-60 to collect slowly the most purulent portion of each sputum sample. Rub the end of the tip on the bottom of the cup to avoid and cut strings.
- Transfer 60 µl of the sputum.
- Transfer the sample slowly into the heating tube. Slowly rinse the tip once to remove the sputum.
- Mix the contents of the tube by shaking.
- Incubate the tube in the HumaLoop T heating unit at 90°C for 5 min.

2. Loopamp™ PURE DNA extraction

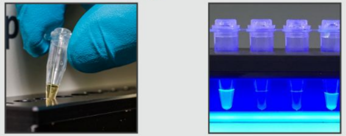
- Remove the cap of the adsorbent tube but do not discard it.
- Screw the heating tube onto the adsorbent tube.
- Mix the lysed sample with the powder in the adsorbent tube by shaking thoroughly.
- Shake the tube until a milky solution is obtained.
- Screw the injection cap onto the other side of the adsorbent tube.
- Extract 30 µl of the DNA directly into the reaction tube by squeezing the adsorbent tube.

3. Loop-mediated isothermal amplification



- Incubate the tube upside down for 2 min (use timer) at room temperature to reconstitute the reagents in the cap.
- Mix the contents of the tube by inverting five times.
- Flick down the reaction tube until the reaction mixture is collected at the bottom.
- Incubate the reaction tube at 67°C for 40 min in the reaction unit. The reaction is automatically inactivated by a further incubation step at 80°C for 5 min.

4. Result reading



- Insert the tube into the detection unit and turn on the UV light.
- Positive results fluoresce green.

A visualized workflow is shown. Please always refer to the latest versions of the Instructions for Use.

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